BTS-V(S)-05-25-5149 Reg. No. | | | | | | | | |

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B.Tech. Degree V Semester Supplementary Examination May 2025

CE/CS/EC/EE//IT/ME/SE 19-200-0501 NUMERICAL AND STATISTICAL METHODS

(2019 Scheme)

Time: 3 Hours Maximum Marks: 60

SCHOOL OF ENGINEERING, KOCHI
COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY
LIBRARY 2022

Course Outcomes
On successful completion of the course, the students will be able to:

CO1: Solve algebraic and transcendental equations by numerical methods.
CO2: Solve numerical differentiation and integration problems.
CO3: Compute the mean and variance of a probability distribution including the binomial distribution.
CO4: Test hypotheses on data.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create
PO – Programme Outcome

PART A

(Answer *ALL* questions)

			(8 × 3 = 24)	Marks	BL	CO	PO
I.	(a)	Find the value of $\frac{1}{31}$ correct to 4 decimals using Newton Raphson method.		3	L2	1	2
	(b)	Construct the backward difference table for $f(x)=x^3-3x^2-5x-7$, for $x=-1, 0, 1, 2, 3, 4, 5$.		3	L2	1	1
	(c)	Evaluate by Taylors series method $y(0,2)$, given $\frac{dy}{dx}=x^2y-1$ and $y(0)=1$.		3	L2	2	1
	(d)	Find $\int_0^1 \frac{1}{(1+x^2)}$ by using simpson's 1/3 rule.		3	L3	2	2
	(e)	A Random variable X has the following probability mass function.		3	L4	3	1
		X: -2, -1, 0, 1, 2, 3 P(X): 0.1, k, 0.2, 2k, 0.3, k Find "k" and mean.					
	(f)	If X follows the Poisson law such that $P(X=1)=P(X=3)$, find $P(X=4)$.		3	L2	3	3
	(g)	Define Critical region and level of significance.		3	L2	1	2
	(h)	A Sample of 20 items has mean 42 units and S.D 5 units. Test the hypothesis that it is a random sample from a normal population with mean 45 units.		3	L3	4	1

PART B

(4 × 12 = 48)

II.	(a)	Solve by Gauss–Seidel method $2x+y+6z=9$; $8x+3y+2z=13$; $x+5y+z=7$.	6	L3	1	1
	(b)	Using Regula Falsi method, find the real root of $2x-\log x=7$ correct to 3 decimals.	6	L2	1	2

OR

(P.T.O.)

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			Marks	BL	CO	PO
III.	(a)	Using Newton's Interpolation formula, find y $y(8)$ from the following table.	6	L3	1	2

x:	0	5	10	15	20	25
y:	7	11	14	18	24	32

					Marks	BL	CO	PO
	(b)	Find the polynomial $f(10)$ from the table using Lagrange's formula.			6	L3	1	2

x:	5	6	9	11
y:	12	13	14	16

			Marks	BL	CO	PO
IV.	(a)	Find the Velocity and acceleration at $t = 0.3$ seconds given	6	L2	2	3

Time(t):	0	0.1	0.2	0.3	0.4	0.5	0.6
Distance(x):	30.13	31.62	32.87	33.64	33.95	33.81	33.24

			Marks	BL	CO	PO
	(b)	Given $\frac{dy}{dx} = x + y$, $h = 0.1$ and $y(0) = 1$; find $y(0.5)$ by Euler's Method.	6	L4	2	1

OR

			Marks	BL	CO	PO
V.	(a)	Apply Runge-Kutta method of 4th order to solve $y(0.2)$ in steps of 0.1, if $\frac{dy}{dx} = x + y^2$ and $y(0) = 1$.	8	L4	2	1
	(b)	Prove that $hD = -\log(1 - \nabla)$.	4	L3	2	3
VI.	(a)	Derive mean and variance of binomial distribution.	6	L1	3	2
	(b)	6 coins are tossed 6400 times. Use poisson distribution to find the probability of getting 6 heads 'x' times.	6	L5	3	1

OR

			Marks	BL	CO	PO
VII.	(a)	In a normal distribution, 31% of items are under 45 and 8% are over 64. Find mean and standard deviation of the distribution.	6	L3	3	3
	(b)	Fit a straight line given	6	L5	3	1

x:	0	1	2	3	4
y:	1	1.8	3.3	4.5	6.3

			Marks	BL	CO	PO
VIII.	(a)	The average marks in Maths of a sample of 100 students was 51 with a standard deviation of 6 marks. Could this have been a random sample from a population with average marks 50?	6	L3	4	4
	(b)	The two random samples reveal the following data.	6	L3	4	4

Sample Number	Size	Mean	Variance
I	16	440	40
II	25	460	42

Test whether the sample come from same normal population at 5% level of significance.

OR

			Marks	BL	CO	PO
IX.	(a)	In two independent samples of sizes 8 and10 and the sum of squares of deviations from sample means were 84.4 and 102.6. Test whether the difference in variances of population is significant or not.	6	L4	4	3
	(b)	The standard deviation of 20 observations from a normal population was found to be 5. Examine whether the sample was taken from a population with standard deviation5.3. Apply chi-square test.	6	L3	4	4

Bloom's Taxonomy Levels
L1 – 8.33%, L2 – 29.17%, L3 – 33.33%, L4 – 16.67%, L5 – 12.5%.

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